Test Series No.

04

NEET Success Kit [2026-27]

DURATION: 195 Minutes | **M. MARKS: 720**

DATE: 10/06/2026

SYLLABUS

Physics:	Full Syllabus
Chemistry:	Full Syllabus
Botany:	Full Syllabus
Zoology:	Full Syllabus

GENERAL INSTRUCTIONS:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours 15 minutes** duration and the Test Booklet contains **180** multiple choice questions (four options with a single correct answer).
3. There is only one correct response for each question.
4. Each **correct** answer will give **4 Mark** while **1 Mark** will be **deducted** for a **wrong** answer. The maximum marks are 720.
5. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR INSTRUCTIONS:

1. Use only blue/black dark ballpoint pen.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS): _____

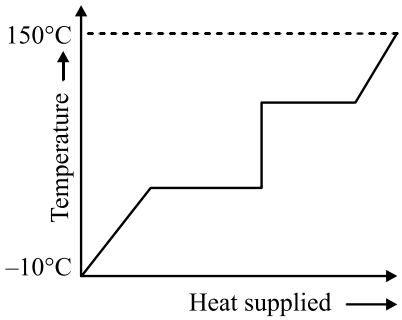
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OMR Bar Code Number: _____

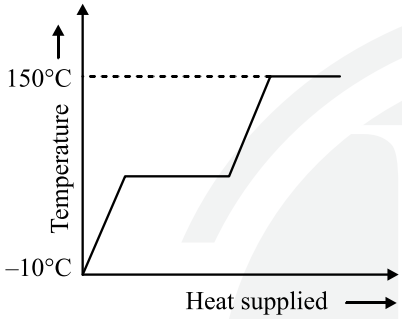
Student's Signature: _____ **Invigilator's Signature:** _____

1. A block of ice at –10°C is slowly heated and converted to steam at 150°C at 1 atm pressure. Which of the following curves most appropriately represent the phenomenon?

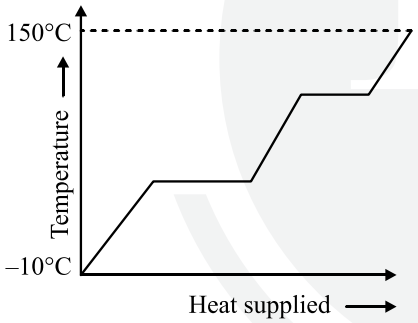
(1)



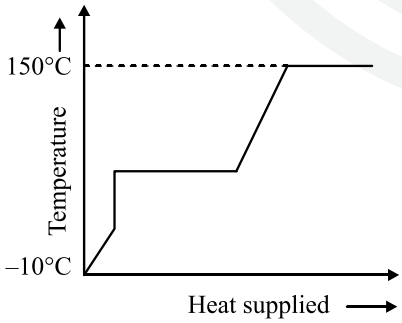
(2)



(3)



(4)



2. A nucleus of mass number 35 splits into two nuclei having mass number 27 and 8. The ratio of radii of two daughter nuclei respectively is:

(1) 27 : 8

(2) 1 : 1

(3) 4 : 3

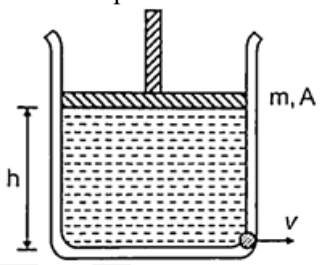
(4) 3 : 2

3. A horizontal force of 10 N is applied on a body of mass 20 kg which is at rest on a smooth horizontal surface at time $t = 0$. The work done by the force in 3rd second is:

(1) 12.5 J (2) 7.5 J

(3) 14 J (4) 20 J

4. A cylindrical vessel contains a liquid of density ρ upto a height h. The liquid is closed by a piston of mass m and area of cross-section A. There is a small hole at the bottom of the vessel. The speed v with which the liquid comes out of the hole is:



(1) $\sqrt{2gh}$ (2) $\sqrt{2\left(gh+\frac{mg}{\rho A}\right)}$

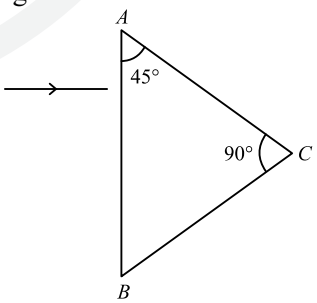
(3) $\sqrt{2\left(gh+\frac{mg}{A}\right)}$ (4) $\sqrt{2gh+\frac{mg}{A}}$

5. What is the dimensional formula of $[\varepsilon_0 G]$ where ε_0 is the permittivity of free space and G is universal gravitational constant?

(1) Dimensionless (2) $[M^{-2}A^2T^2]$

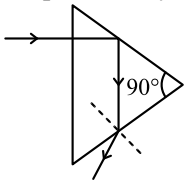
(3) $[M^2A^{-2}T^{-2}]$ (4) $[M^{-1}A^2\mathrm{T}]$

6. A ray of light is directed in front of a right angled prism ABC. The prism is made of crown glass with critical angle 41°.

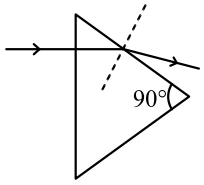


If a ray is incident perpendicular to face AB, then the **correct** path of ray will be:

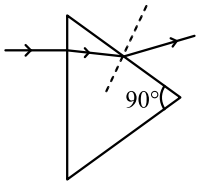
(1)



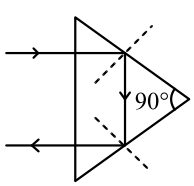
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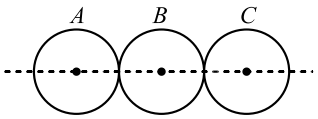
(3)



(4)



7. The masses of three spheres A, B and C having same radius R are M, M and $2M$ respectively. They are placed touching each other as shown. The distance of their centre of mass from centre of sphere C is:



(1) $1.5R$ (2) $2.5R$

(3) R (4) $3.5R$

8. A straight long infinite wire carrying 12 A current is placed along the y-axis. What is the magnetic field at a point with coordinates (3m, 4m)?

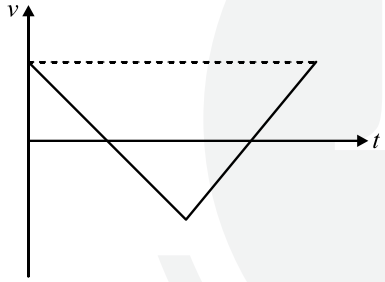
(1) 2×10^{-7} T

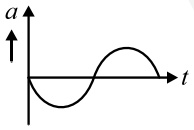
(2) 4×10^{-7} T

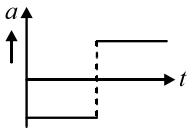
(3) 8×10^{-7} T

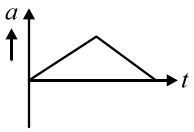
(4) 6×10^{-7} T

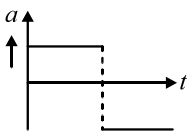
9. The velocity-time (v-t) graph of a particle moving in straight line is shown in figure. The corresponding acceleration-time (a-t) graph is:



(1) 

(2) 

(3) 

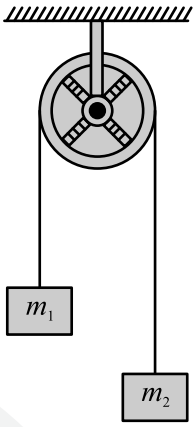
(4) 

10. An uncharged capacitor of capacitance 4 μF is connected with a battery of 10 volt. The energy produced as heat energy in connecting wires till the capacitor is completely charged would be:

(1) 200 μJ (2) 300 μJ

(3) 600 μJ (4) 100 μJ

11. A light string passing over a smooth light fixed pulley connects two blocks of masses m_1 and m_2. If the acceleration of the each block is $\frac{5}{4}$ m/s^2, then the ratio of masses $\left(\frac{m_1}{m_2}\right)$ is ($m_1 > m_2$, $g = 10$ m/s^2)



(1) $\frac{7}{4}$ (2) $\frac{4}{3}$

(3) $\frac{11}{9}$ (4) $\frac{9}{7}$

12. The electric field in a region is given by $\vec{E} = \left(\frac{3}{5}\hat{i} + \frac{6}{5}\hat{j}\right) \times 10^3$ N/C. The electric flux due to this field through a rectangular surface of area 0.2 m^2 whose plane is parallel to the X-Z plane is:

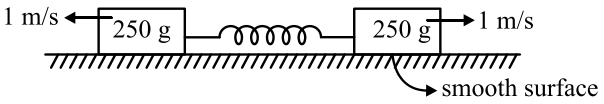
(1) 120 Nm^2/C

(2) 160 Nm^2/C

(3) 180 Nm^2/C

(4) 240 Nm^2/C

13. Two blocks each of mass 250 gram are connected to spring in its natural length and having spring constant 2 Nm^{-1}. If both are given velocity 1 m/s in opposite directions, then maximum elongation of the spring is:



(1) $\frac{1}{2\sqrt{2}}$ m

(2) $\frac{1}{2}$ m

(3) $\frac{1}{4}$ m

(4) $\frac{1}{\sqrt{2}}$ m

[Contd...

14. Given below are two statements:

Statement I: Photosensitivity of a metal is high if its work function is small.

Statement II: If v_0 is the incident threshold frequency, then work function of metal surface is hv_0^2 (h is Planck's constant)

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Statement I and Statement II both are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Statement I and Statement II both are incorrect.

15. Given below are two statements:

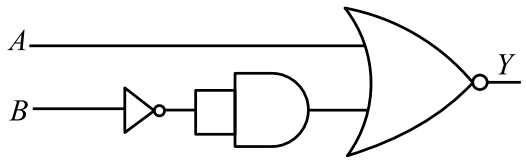
Statement I: In isothermal expansion during a carnot cycle for a heat engine, heat is released by the system.

Statement II: If the temperature of the source is 100°C and that of the sink is 50°C, then the efficiency of an engine working on Carnot cycle is 50%.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Statement I and Statement II both are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Statement I and Statement II both are incorrect.

16. A combination of gates is shown below. Which of the options given below represent the **correct** Boolean expression of the output Y.



(1) $Y = A + \overline{B}$ (2) $Y = \overline{A} + B$
(3) $Y = A \cdot \overline{B}$ (4) $Y = \overline{A} \cdot B$

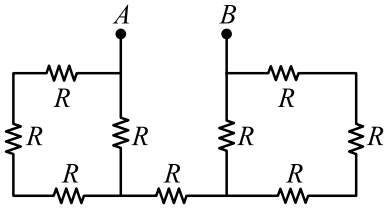
17. The average emf induced in a coil in which the current changes from 2 ampere to 4 ampere in the same direction in 0.02 s is 8 volt. What is the self inductance of the coil?

(1) 0.04 H (2) 0.08 H
(3) 0.4 H (4) 0.8 H

18. Binding energy of a certain nucleus is 18×10^8 J. How much is the difference between total mass of all the nucleons and nuclear mass of the given nucleus?

(1) 20 μg (2) 2 μg
(3) 0.2 μg (4) 10 μg

19. The equivalent resistance of the circuit across A and B as shown in the figure is;



(1) $\frac{5R}{2}$ (2) $\frac{3R}{4}$
(3) $\frac{5R}{3}$ (4) $\frac{2R}{5}$

20. A potential difference of 20V is applied across a conductor of length 10 cm. If drift velocity of electrons is 2×10^{-4} m/s, then electron mobility will be:

(1) 1×10^{-6} m^2 V^{-1} s^{-1}
(2) 2×10^{-6} m^2 V^{-1} s^{-1}
(3) 3×10^{-6} m^2 V^{-1} s^{-1}
(4) 4×10^{-6} m^2 V^{-1} s^{-1}

21. What will be the effect on the root mean square (RMS) velocity of oxygen molecules, if the temperature is doubled and oxygen molecule dissociates into atomic oxygen?

(1) RMS velocity of atomic oxygen remain same as RMS velocity of oxygen molecule.
(2) RMS velocity of atomic oxygen become half of RMS velocity of oxygen molecule.
(3) RMS velocity of atomic oxygen become two times the RMS velocity of oxygen molecule.
(4) RMS velocity of atomic oxygen become four times the RMS velocity of oxygen molecule.

22. Calculate the de Broglie wavelength of electron having kinetic energy 3.2×10^{-17} J (mass of electron $= 1 \times 10^{-30}$ kg, $h = 6.6 \times 10^{-34}$ Js)

(1) 9.60×10^{-11} m
(2) 8.25×10^{-11} m
(3) 6.25×10^{-11} m
(4) 5.00×10^{-11} m

[Contd...

23. A ball is projected at an angle θ with vertical from ground. Value of θ for which horizontal range and maximum height attained by ball are equal is:

(1) $\tan^{-1}(4)$

(2) $\tan^{-1}(2)$

(3) $\cot^{-1}(2)$

(4) $\cot^{-1}(4)$

24. The voltage between the plates of a parallel plate capacitor is changing at the rate of 10 V/s. If the displacement current in the capacitor is 2×10^{-5} A, then the capacitance of capacitor is:

(1) 4 μF

(2) 3 μF

(3) 1 μF

(4) 2 μF

25. The amplitude of a particle in SHM is 5 cm and its time period is π second. At a displacement of 3 cm from its mean position, the velocity (in cm/sec) will be;

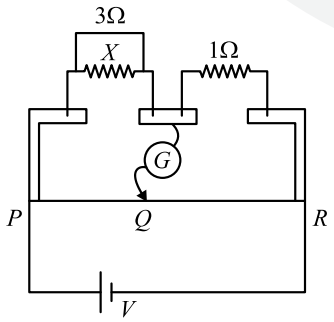
(1) 8

(2) 12

(3) 2

(4) 16

26. In the meter bridge, galvanometer shows zero deflection if the balancing length $PQ = 40$ cm. The value of unknown resistance X is;



(1) $\frac{6}{7}\Omega$

(2) $\frac{7}{6}\Omega$

(3) $\frac{3}{2}\Omega$

(4) $\frac{2}{3}\Omega$

27. Variation of density of a solid with change in temperature is given as $\rho = \left(\frac{4.2}{2 + 3.6 \times 10^{-5} \Delta t}\right)$, where Δt is in °C and ρ is in g/cm^3. Coefficient of volume expansion of the solid is:

(1) 3.6×10^{-5} /°C

(2) 1.8×10^{-5} /°C

(3) 3.6×10^{5} /°C

(4) 1.8×10^{5} /°C

28. A bomb initially at rest explodes by itself into three equal mass fragments. The velocities of two fragments are $(3\hat{i} + 2\hat{j})$ m/s and $(-\hat{i} - 4\hat{j})$ m/s. The velocity of the third fragment is (in m/s):

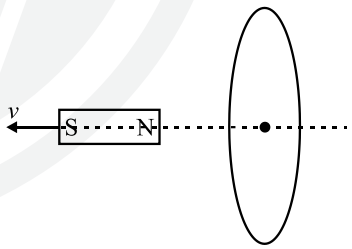
(1) $2\hat{i} + 2\hat{j}$

(2) $2\hat{i} - 2\hat{j}$

(3) $-2\hat{i} + 2\hat{j}$

(4) $-2\hat{i} - 2\hat{j}$

29. Consider a bar magnet placed along the axis of circular loop of radius r as shown in figure. Due to the bar magnet being pulled away from loop along the axis with constant speed v, an emf is induced. Magnitude of induced emf in this circular loop will:



(1) decrease with time

(2) increase with time

(3) be zero

(4) be constant (non-zero) with time.

30. The ratio of relative permeability μ_r and magnetic susceptibility χ of magnetic material is:

(1) $\frac{1}{\chi} + 1$

(2) $\chi + 1$

(3) $\frac{\mu_r}{\mu_r - 1}$

(4) Both (1) and (3)

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31. When a satellite is shifted from farther orbit to a nearer orbit, its

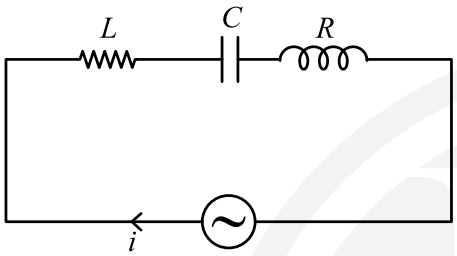
(1) gravitational potential energy increases and kinetic energy decreases.

(2) gravitational potential energy decreases and kinetic energy increases.

(3) both gravitational potential energy and kinetic energy decreases.

(4) both gravitational potential energy and kinetic energy increases.

32. Choose the **correct** statements for series LCR circuit as shown in figure.



A. Voltage across L and C differ by $\frac{\pi}{2}$ radian phase angle.

B. Current across C and R are in same phase

C. Voltage across C and current through L differ by $\frac{\pi}{2}$ radian phase angle.

(1) B only

(2) B and C only

(3) A and C only

(4) A, B and C

33. A second's pendulum gives correct time period on the surface of moon. What will be its time period on the earth's surface?

(1) 2 s (2) $2\sqrt{6}$ s

(3) $\frac{2}{\sqrt{6}}$ s (4) $6\sqrt{2}$ s

34. A magnet having magnetic moment $\vec{M}=(10\hat{i})\,\text{Am}^2$ is placed in a magnetic field $\vec{B}=(2\hat{i}+3\hat{j})$ T. The torque acting on the magnet is:

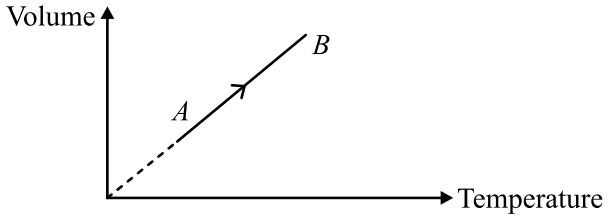
(1) $(20\hat{j}+30\hat{k})$ N-m

(2) $30\hat{k}$ N-m

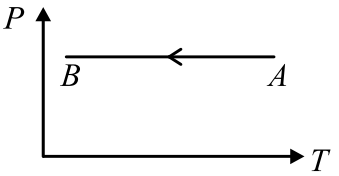
(3) $(30\hat{j}+20\hat{k})$ N-m

(4) $20\hat{k}$ N-m

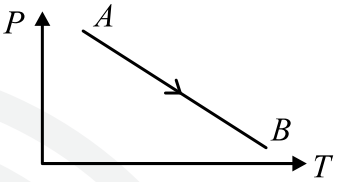
35. An ideal gas undergoes a thermodynamic process as shown in figure. The corresponding Pressure-Temperature (P-T) graph is:



(1)



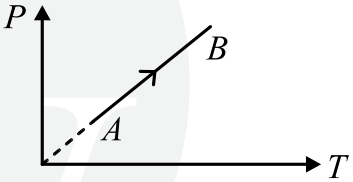
(2)



(3)



(4)

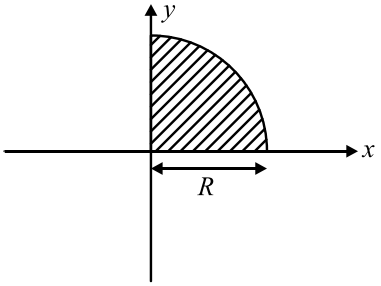


36. In Young's double slit experiment, 60 fringes are observed in the central view zone with light of wavelength 6000 Å. The number of fringes that will be observed in the same view zone with the light of wavelength 4000 Å is;

(1) 40 (2) 60

(3) 90 (4) 70

37. The coordinates of centre of mass of uniform quarter disc of radius R as shown in figure is:



(1) $\frac{2R}{\pi}, \frac{4R}{3\pi}$ (2) $\frac{2R}{\pi}, \frac{2R}{\pi}$

(3) $\frac{4R}{3\pi}, \frac{4R}{3\pi}$ (4) $\frac{4R}{3\pi}, \frac{2R}{\pi}$

[Contd...

38. As per Bohr model, the minimum energy required to remove an electron from the ground state of doubly ionized Li atom is;

(1) 1.51 eV

(2) 13.6 eV

(3) 40.8 eV

(4) 122.4 eV

39. Standing waves are produced in a string fixed at both ends. What are the total number of antinodes if the string is vibrating in 5th overtone?

(1) 6

(2) 7

(3) 5

(4) 8

40. At what distance on the axis from the centre of a circular current carrying coil of radius r, the magnitude of magnetic field becomes $\frac{1}{8}$th of the magnitude of magnetic field at centre?

(1) $\sqrt{2}r$

(2) $2^{3/2}r$

(3) $\sqrt{3}r$

(4) $3\sqrt{2}r$

41. Match **List-I** (physical quantities) with **List-II** (their corresponding units).

List-I		List-II	
(A)	Rydberg constant	(I)	kg m^{-1} s^{-1}
(B)	Planck's constant	(II)	kg m^2 s^{-1}
(C)	Magnetic field energy density	(III)	m^{-1}
(D)	Coefficient of viscosity	(IV)	kg m^{-1} s^{-2}

(1) A-III, B-II, C-IV, D-I

(2) A-III, B-IV, C-II, D-I

(3) A-II, B-III, C-I, D-IV

(4) A-II, B-I, C-III, D-IV

42. A container of depth $2H$ is filled with two immiscible transparent liquids of refractive index μ_1 and μ_2 respectively. The depth of each liquid is H. When viewed normally from above, the apparent depth of the vessel is;

(1) $\frac{H}{\mu_1}+\frac{H}{\mu_2}$ (2) $\frac{H}{\mu_1}-\frac{H}{\mu_2}$

(3) $\frac{2H}{\mu_1}+\frac{2H}{\mu_2}$ (4) $\frac{H}{2\mu_1}-\frac{H}{\mu_2}$

43. The equation of a transverse wave is given by $y = 10 \sin \pi (0.01\, x - 2.00\, t)$ where y and x are in cm and t in second. The speed of the wave will be:

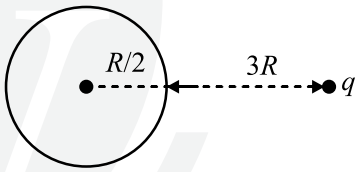
(1) 100 cm/s

(2) 200 cm/s

(3) 50 cm/s

(4) 150 cm/s

44. A positive point charge q is placed at distance 3R from the surface of a metallic shell of radius $\frac{R}{2}$. The potential at the centre of shell due to induced charge on shell is:



(1) $\frac{kq}{3R}$ (2) $\frac{-2kq}{7R}$

(3) $\frac{-3kq}{5R}$ (4) 0

45. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Steel is used in the construction of buildings and bridges.

Reason R: Steel is elastic and its elastic limit is high.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.

(2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

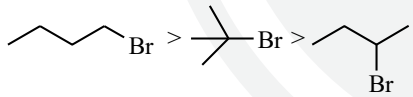
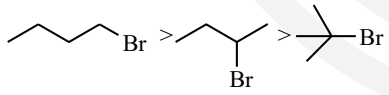
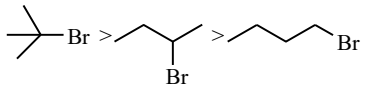
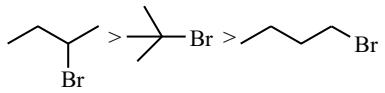
(4) Both A and R are true but R is NOT the correct explanation of A.

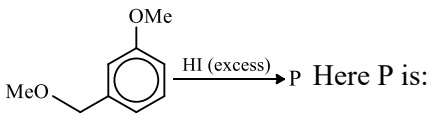
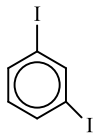
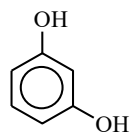
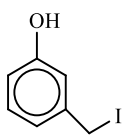
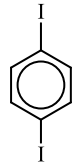
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46. The extra stability of half-filled subshell is due to:
A. Symmetrical distribution of electrons
B. Larger exchange energy
C. Lower electron-electron repulsion
D. Presence of paired electrons in the same orbital
Identify the **correct** statements.
(1) A, B and C only (2) A, C and D only
(3) B and D only (4) A, B, C and D

47. What is the number of photons of light with a wavelength of 6630 pm that provide 1 J of energy? (Given: $h = 6.63 \times 10^{-34}$ Js)
(1) 3.33×10^{16} (2) 3.33×10^{17}
(3) 3.33×10^{15} (4) 3.33×10^{18}

48. The **correct** statements from the following are:
A. The order of ligands in the spectrochemical series is completely explained by crystal field theory.
B. Valence bond theory does not give a quantitative interpretation of kinetic stability of coordination compounds.
C. The hybridisation involved in $[Ni(CN)_4]^{2-}$ is dsp^2.
D. The number of possible stereoisomers of cis-$[CoCl_2(en)_2]^+$ is one.
Choose the **correct** answer from the options given below:
(1) A, D only (2) A, C only
(3) B, D only (4) B, C only

49. The **correct** order of boiling point for the given compounds is:
(1) 
(2) 
(3) 
(4) 

50.  Here P is:
(1)  (2) 
(3)  (4) 

51. An unknown compound (A) has molecular formula C_5H_8. When treated with excess Br_2, it forms a substance with formula $C_5H_8Br_4$. Compound (A) gives a white precipitate with ammoniacal silver nitrate. Compound (A) may be:
(1) Pent-1-yne
(2) Pent-2-yne
(3) Pent-1-ene
(4) Cyclopentene

52. In which of the following processes, entropy increases?
(1) $C(s) + O_2(g) \rightarrow CO_2(g)$
(2) $Fe_2O_3(s) + 3H_2(g) \rightarrow 2Fe(s) + 3H_2O(\ell)$
(3) $N_2O_4(g) \rightarrow 2NO_2(g)$
(4) $H_2O(vap) \rightarrow H_2O(\ell)$

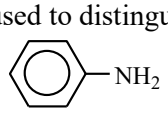
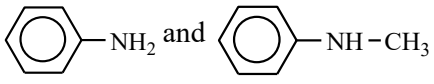
53. The solubility of Ag_2CO_3 in water at 298 K is:
(Given: K_{sp} of Ag_2CO_3 is 8×10^{-12})
(1) $(2)^{1/3} \times 10^{-4}$ M
(2) 2×10^{-4} M
(3) $(3)^2 \times 10^{-4}$ M
(4) 3×10^{-4} M

54. Match the List-I with List-II.

List-I		List-II	
(A)	B_2	(I)	4 electrons in bonding molecular orbitals
(B)	C_2	(II)	Paramagnetic
(C)	N_2	(III)	No σ bond
(D)	Li_2	(IV)	Bond order is three

Choose the **correct** answer from the options given below;
(1) A-II, B-I, C-IV, D-III
(2) A-I, B-III, C-II, D-IV
(3) A-III, B-IV, C-I, D-II
(4) A-II, B-III, C-IV, D-I

55. All of the given oxides are amphoteric **except**:
(1) PbO
(2) SnO_2
(3) Tl_2O_3
(4) Al_2O_3

56. Isocyanide test is used to distinguish between:
(1) $C_2H_5NH_2$ and 
(2) $(CH_3)_2NH$ and $(CH_3)_3N$
(3) 
(4) CH_3NH_2 and $C_2H_5NH_2$

[Contd...

57. Match **List-I** with **List-II**:

List-I		List-II	
(A)	Glycosidic linkage	(I)	2° and 3° structures of proteins
(B)	Peptide linkage	(II)	Amide linkage between –COOH and $–NH_2$ group
(C)	Disulphide linkage	(III)	Linkage between nucleotides
(D)	Phosphodiester linkage	(IV)	Linkage between two monosaccharides

Choose the **correct** answer from the options given below:

(1) A-IV, B-II, C-I, D-III
(2) A-II, B-IV, C-I, D-III
(3) A-IV, B-III, C-II, D-I
(4) A-I, B-II, C-III, D-IV

58. The **incorrect** statement among the following is:

(1) PCl_5 exists but NCl_5 does not.
(2) SO_2 can act as both oxidising and reducing agent.
(3) PH_3 has lower proton affinity than NH_3.
(4) NO can dimerise easily.

59. For the redox reaction,

$$Fe^{2+} + Cr_2O_7^{2-} + H^+ \rightarrow Fe^{3+} + Cr^{3+} + H_2O$$

The **correct** coefficients of the reactants for the balanced equation are:

	Fe^{2+}	$Cr_2O_7^{2-}$	H^+
(1)	6	2	7
(2)	6	1	14
(3)	4	3	7
(4)	2	1	12

60. Which of the following alkenes give acetone as one of the product on reductive ozonolysis?

(1) 3-Methylbut-1-ene
(2) 2-Methylprop-1-ene
(3) 3-Methylpent-2-ene
(4) 3-Ethylpent-1-ene

61. Benzene has uniform C–C bond distances of:

(1) 120 pm (2) 134 pm
(3) 139 pm (4) 154 pm

62. Which among the following is expected to be most stable?

(1) $CH_3CH_2O^-$ (2) $FCH_2CH_2O^-$
(3) $O_2NCH_2CH_2O^-$ (4) $NCCH_2CH_2O^-$

63. If N and S both are present in organic compound, then during Lassaigne's test, these elements are converted into:

(1) NaCN (2) NaSCN
(3) NH_2CSNH_2 (4) N_2S_3

64. Given below are two statements:

Statement I: Solubility of gases in water increases with decrease of temperature.

Statement II: Vapour pressure is a colligative property.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

65. If Λ_m^0 for NaCl, HCl and CH_3COONa are x, y and z S cm^2 mol^{-1} respectively then Λ_m^0 (in S cm^2 mol^{-1}) for CH_3COOH is:

(1) z + x – y (2) x + y – z
(3) x – y – z (4) z + y – x

66. For the reactions:

$$Sn^{2+} + 2e^- \rightarrow Sn$$

$$Sn^{4+} + 4e^- \rightarrow Sn$$

The standard electrode potentials are:

$E^\circ_{Sn^{2+}/Sn} = -0.140$ V and $E^\circ_{Sn^{4+}/Sn} = 0.010$ V

The standard oxidation potential for $Sn^{2+} \rightarrow Sn^{4+} + 2e^-$ is:

(1) +0.160 V (2) -0.010 V
(3) -0.160 V (4) +0.140 V

67. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Actinoid contraction is greater from element to element than lanthanoid contraction.

Reason R: In actinoids, last election enters in 4f subshell.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

[Contd...

68. The **correct** order of increasing field strength of given ligands is:
(1) $Cl^- < NH_3 < HO^- < en$
(2) $HO^- < Cl^- < en < NH_3$
(3) $Cl^- < HO^- < NH_3 < en$
(4) $NH_3 < en < Cl^- < HO^-$

69. The formula of pyrosulphuric acid is:
(1) $H_2S_2O_7$ (2) $H_2S_2O_6$
(3) $H_2S_2O_8$ (4) $H_2S_2O_5$

70. One mole of a non ideal gas changes state from (2 atm, 4 L, 250 K) to (3 atm, 6 L, 300 K) with a change in internal energy (ΔU) = 50 L atm. The change in enthalpy of the process is:
(1) 60 L atm (2) 50 L atm
(3) 65 L atm (4) 55 L atm

71. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Molality of a solution does not change with temperature.
Reason R: Mass remains unaffected with temperature.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

72. What is the lowest value of n that allows g orbitals to exist?
(1) 4 (2) 3
(3) 6 (4) 5

73. The largest and smallest size among Mg, Mg^{2+}, Al and Al^{3+} respectively are:
(1) Al and Mg^{2+} (2) Mg and Al^{3+}
(3) Mg and Mg^{2+} (4) Al and Al^{3+}

74. Which one of the following statements is incorrect related to Molecular Orbital Theory?
(1) The σ* antibonding molecular orbital has a node between the nuclei.
(2) In the formation of a bonding molecular orbital, the electron waves of combining atoms reinforce each other.
(3) Molecular orbitals formed from $2p_x$ and $2p_y$ orbitals are symmetrical around the bond axis.
(4) A π-bonding molecular orbital has greater electron density above and below the internuclear axis.

75. For the reaction:
$SO_2Cl_2(g) \rightleftharpoons SO_2(g) + Cl_2(g)$,
K_p/K_c is equals to:
(1) $(RT)^{-2}$ (2) $(RT)^2$
(3) RT (4) $(RT)^{-1}$

76. Given the standard electrode potentials,
$Sn^{2+}/Sn = -0.14$ V, $Ag^+/Ag = 0.80$ V,
$Al^{3+}/Al = -1.66$ V, $Cu^{2+}/Cu = 0.34$ V
The maximum reducing power is of:
(1) Sn (2) Ag
(3) Al (4) Cu

77. Given below are two statements:
Statement I: Aluminium is the most abundant metal on earth crust.
Statement II: CCl_4 is easily hydrolysed by water.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

78. Identify the product (D) in the following sequence of reactions:

$$CH_3COOH \underset{\text{(aq. solution)}}{\xrightarrow{NaOH}} A \xrightarrow{\text{electrolysis}} B \xrightarrow{Cl_2,\ hv} C \xrightarrow[\Delta]{\text{alc.KOH}} D$$

(1) C_2H_6 (2) C_2H_4
(3) C_6H_6 (4) CH_4

79. The osmotic pressure associated with the fluid inside the blood cell is equivalent to that of:
(1) 0.3% (mass/volume) NaCl solution
(2) 0.5% (mass/volume) NaCl solution
(3) 0.9% (mass/volume) NaCl solution
(4) 0.7% (mass/volume) NaCl solution

80. What mass of Al is deposited by passing a current of 9.65 amperes for 100 seconds through molten Al_2O_3?
(1) 0.06 g (2) 0.09 g
(3) 0.03 g (4) 0.01 g

81. Order of a reaction whose half life becomes double on doubling the concentration of reactant, is:
(1) 0 (2) 3
(3) 1 (4) 2

[Contd...

82. Give below are two statements:

Statement-I: Noble gases have very low boiling points.

Statement-II: Noble gases are monoatomic and are held together only by weak dispersion forces. Therefore they liquefy only at very low temperatures.

In the light of the above statements, choose the correct answer from the options given below:

(1) Statement-I is false but Statement-II is true
(2) Both Statement-I and Statement-II are true
(3) Statement-I is true but Statement-II is false
(4) Both Statement-I and Statement-II are false

83. Match the List-I with List-II.

List-I (Process)		List-II (Catalyst)	
(A)	Oxidation of SO_2 into SO_3	(I)	$TiCl_4$ with $Al(CH_3)_3$
(B)	Manufacture of polythene	(II)	Iron
(C)	Wacker process	(III)	$PdCl_2$
(D)	Haber process	(IV)	V_2O_5

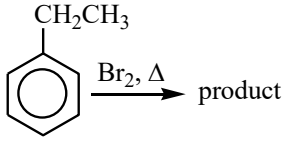
Choose the **correct** answer from the options given below;

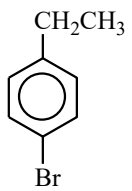
(1) A-IV, B-I, C-III, D-II
(2) A-I, B-III, C-II, D-IV
(3) A-III, B-IV, C-I, D-II
(4) A-II, B-III, C-IV, D-I

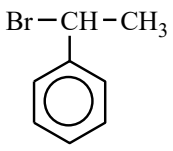
84. Octacarbonyldicobalt (0) has a Co–Co bond bridge by:

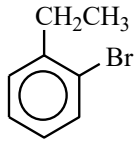
(1) one CO group
(2) two CO groups
(3) three CO groups
(4) four CO group

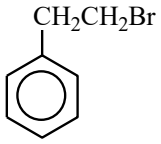
85. Major product of the following reaction is:



(1) 

(2) 

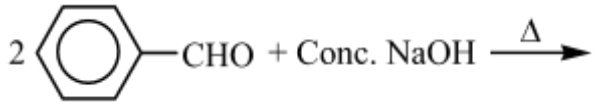
(3) 

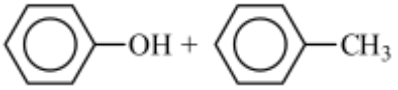
(4) 

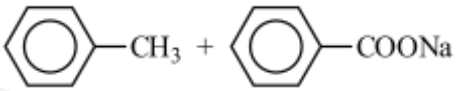
86. The commercial alcohol is made unfit for drinking by mixing in it:

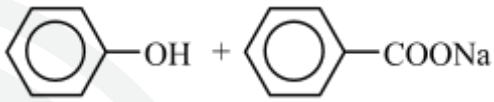
(1) Zinc carbonate and aniline
(2) Copper sulphate and pyridine
(3) Magnesium oxide and furan
(4) Ferric chloride and thiophene

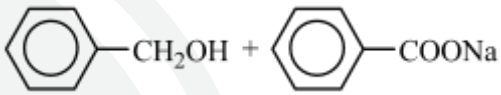
87. The products formed in the given reaction is;



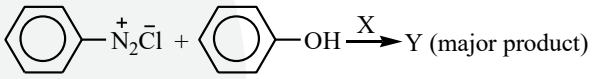
(1) 

(2) 

(3) 

(4) 

88. X and Y in the following reaction respectively are:



(1) OH^- and yellow dye
(2) H^+ and yellow dye
(3) OH^- and orange dye
(4) H^+ and orange dye

89. Match the **List-I** with **List-II**.

List-I		List-II	
(A)	Fibrous protein	(I)	Glucagon
(B)	Essential amino acid	(II)	Riboflavin
(C)	Hormone	(III)	Myosin
(D)	Vitamin B_2	(IV)	Histidine

Choose the **correct** answer from the options given below;

(1) A-III, B-IV, C-I, D-II
(2) A-III, B-IV, C-II, D-I
(3) A-II, B-I, C-III, D-IV
(4) A-II, B-III, C-IV, D-I

90. Compound X on reaction with O_3 followed by Zn/H_2O gives methanal and 2-methylpropanal as products. Compound X is:

(1) 2-Methylbut-2-ene
(2) 3-Methylbut-1-ene
(3) Pent-2-ene
(4) 2-Methylbut-1-ene

[Contd...

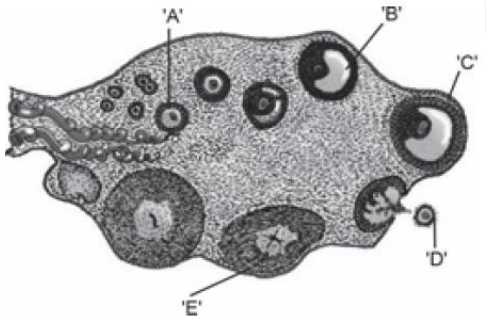
91. "The integration of natural science and organisms, cells, parts thereof, and molecular analogues for products and services". This definition of biotechnology was given by
(1) CDRI, India
(2) WHO, Geneva
(3) EFB, Europe
(4) NACO, India

92. The process by which seedless fruits are produced is known as:
(1) embryogeny. (2) parthenocarpy.
(3) apomixis. (4) polyembryony.

93. In transverse section of root of flowering plants:
(1) the protoxylem lies towards the pith.
(2) arrangement of primary xylem is called exarch type.
(3) the metaxylem lies towards the periphery of the organ.
(4) only metaxylem is present and no protoxylem is found.

94. Select the **correct** option for genera that belong to the same class of algae.
(1) *Sargassum*, *Laminaria* and *Gelidium*
(2) *Ectocarpus*, *Dictyota* and *Laminaria*
(3) *Porphyra*, *Ectocarpus* and *Chara*
(4) *Chara*, *Fucus* and *Polysiphonia*

95. Observe the above given diagram and select the **incorrect** statement w.r.t labellings 'A' to 'E'.



(1) The primary oocyte gets surrounded by a layer of granulosa cell and it is called 'A'
(2) 'C' ruptures to release the 'D' by the process called ovulation
(3) 'B' is represented by a fluid filled cavity called antrum and it possesses primary oocyte arrested in meiosis-II
(4) 'E' is a temporary endocrine structure that releases high amount of progesterone.

96. Which of the following is the most appropriate statement defining infertility?
(1) Couple is unable to produce children inspite of unprotected sexual co-habitation even after two years.
(2) It is the inability to produce a viable o spring and is always due to abnormalities in the female partner.
(3) Infertility is only due to immature sex organs.
(4) Couple is unable to produce children inspite of unprotected sexual co-habitation even after two months.

97. Choose the **false** statement concerning energy flow in Grazing food chain (GFC) and Detritus food chain (DFC).
(1) GFC is major conduit for energy flow in the aquatic ecosystem.
(2) In the DFC first trophic level is formed by producers.
(3) Much larger fraction of energy flows through the DFC in a terrestrial ecosystem.
(4) The DFC may be interconnected with The GFC.

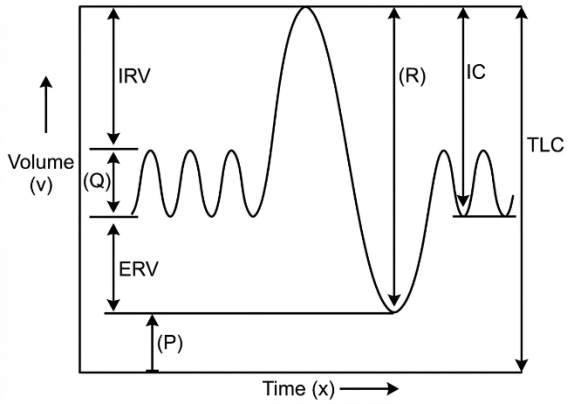
98. Which of the following is **correct** for Photosystem II?
(1) It is not associated with splitting of water.
(2) It is found in membrane or lamellae of the grana only.
(3) It is involved in both cyclic and non-cyclic electron flow.
(4) It is found on the outer surface of the thylakoids.

99. Among the traits studied by Mendel in pea plants, which one was **not** included?
(1) Pod shape and pod colour.
(2) Height of plant and flower position.
(3) Seed shape and seed colour.
(4) Flower shape and leaf colour.

100. The region of the stem where leaves are born are called:
(1) nodes.
(2) internodes.
(3) terminal buds.
(4) axillary buds.

[Contd...

101. Observe the given below diagrammatic representation of pulmonary volumes and capacities of a healthy man.



Select the **correct** option w.r.t. above given graph.

(1) Average numerical value of [ERV + (P)] is equal to the value of [(R) + (P)]

(2) [IRV + (Q)] is equal to 1200 mL

(3) (P) prevents the lungs from collapsing by keeping the alveoli open

(4) (R) cannot be measured by a spirometer.

102. Match the **List-I** with **List-II**.

List-I		List-II	
(A)	Syphilis	(I)	*Haemophilus influenzae*
(B)	Plague	(II)	*Wuchereria malayi*
(C)	Pneumonia	(III)	*Treponema pallidum*
(D)	Elephantiasis	(IV)	*Yersinia pestis*

Choose the **correct** answer from the options given below:

(1) A-I, B-II, C-III, D-IV

(2) A-III, B-IV, C-II, D-I

(3) A-I, B-IV, C-III, D-II

(4) A-III, B-IV, C-I, D-II

103. Select the **correct** option w.r.t. *P. americana*.

(1) Each Malpighian tubule is lined by flattened non -glandular cells with microvilli.

(2) Fat bodies, nephrocytes and urecose glands also help in excretion

(3) A network of trachea open through 10 pairs of spiracles on its dorsal side of the body

(4) Brain is represented by sub-oesophageal ganglion.

104. Which of the following statements is **incorrect**?

(1) In agriculture, there is a method of controlling pests that relies on natural predation rather than introduced chemicals.

(2) A key belief of the organic farmer is that biodiversity furthers health.

(3) Biofertiliser refers to the use of biological methods for controlling plant diseases and pests.

(4) The use of biocontrol measures greatly reduces our dependence on toxic chemicals and pesticides.

105. 'X' hormone is released from posterior pituitary that helps in parturition. Select the **incorrect** statement w.r.t 'X' in relation with parturition.

(1) 'X' along with cortisol and estrogens induces signals for the complex neuroendocrine mechanism required for parturition.

(2) 'X' binds onto the receptors present on endometrium and causes vigorous contractions in uterus.

(3) The 'X' is released from maternal pituitary due to mild uterine contractions caused by foetal ejection reflex.

(4) 'X' secretion is regulated by positive feedback mechanism, where uterine contractions stimulate further release of 'X'.

106. Which of the following is a chemically modified sugar?

(1) Glucosamine

(2) Cellulose

(3) Starch

(4) Glycogen

107. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion (A): In order to force bacteria to take up the plasmid, the bacterial cells are treated with a specific concentration of a divalent cation, such as calcium

Reason (R): In calcium chloride method, the DNA to be transferred is mixed in a solution containing negatively charged divalent ion and the positively charged DNA to form a complex that efficiently passes through the pores present in the plasma membrane of bacteria.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both A and R are true and R is the correct explanation of A.

(2) Both A and R are true but R is not the correct explanation of A.

(3) A is true but R is false.

(4) A is false but R is true.

[Contd...

108. In humans, the first movement of foetus is observed during the

(1) Second month of pregnancy
(2) Fifth month of pregnancy
(3) 1st trimester
(4) 12th week of pregnancy

109. Match the **List-I** with **List-II** and select the **correct** option w.r.t. adult humans.

List-I		List-II	
(A)	Neutrophils	(I)	Are of two major types – 'B' and 'T' forms
(B)	Lymphocytes	(II)	Non-nucleated formed elements that contain respiratory pigment
(C)	Monocytes	(III)	Phagocytic leucocytes that constitute 60–65% of total WBCs
(D)	Erythrocytes	(IV)	Phagocytic agranulocytes that constitute only 6–8% of total WBCs

Choose the **correct** answer from the options given below:

(1) A-IV, B-III, C-II, D-I
(2) A-II, B-I, C-IV, D-III
(3) A-III, B-I, C-IV, D-II
(4) A-III, B-II, C-IV, D-I

110. GM plants are useful in

A. Increasing crop yields
B. Reducing post-harvest losses
C. Increasing reliance on chemical pesticides

Choose the **correct** option.

(1) A and C
(2) B and C
(3) A and B
(4) A, B and C

111. The action of which among the following primarily moves the body to adjust to changes in the environment and to maintain the positions of the various parts of the body?

(1) Epithelial tissue
(2) Muscular tissue
(3) Dense irregular connective tissue
(4) Fluid connective tissue

112. All of the following are true w.r.t codeine, **except**:

(1) It is a polymeric substance-like gum
(2) Belongs to the same category as morphine
(3) It is an alkaloid
(4) It is a secondary metabolite

113. The afferent nerve fibres transmit impulses from tissues/organs to the

(1) Central neural system
(2) Peripheral neural system only
(3) Sympathetic neural system only
(4) Parasympathetic neural system only

114. "All the chordates are vertebrates".

Select the option that **correctly** explains whether the above stated fact is true or false.

(1) True, as all chordates are included in the sub-phylum Vertebrata
(2) False, as protochordates make up the vast majority of the animals in the sub-phylum Vertebrata
(3) True, as both chordates and vertebrates share some basic characteristics like presence of hollow central nervous system
(4) False, as some chordates like tunicates and lancelets lack a vertebral column in their adult stage

115. Choose the **incorrect** statement w.r.t dicot leaf.

(1) Mesophyll is differentiated into palisade and spongy parenchyma.
(2) Mesophyll which possesses chloroplasts carries out photosynthesis.
(3) The size of the vascular bundles is independent of the size of the veins.
(4) The vascular system includes vascular bundles which can be seen in the veins and the midrib.

116. Select the **incorrect** match from the following.

(1) Polysome - mRNA to protein
(2) Plasmid - Antibiotic resistance
(3) Centriole - Spindle fibre
(4) Fimbriae - Locomotion

[Contd...

117. Match the **List-I** with **List-II**.

List-I		List-II	
(A)	Polygenic inheritance	(I)	Down's syndrome
(B)	Sex-linked recessive trait	(II)	Phenylketonuria
(C)	Inborn error of metabolism	(III)	Colour blindness
(D)	Additional copy of 21st chromosome	(IV)	Skin colour in humans

Choose the **correct** answer from the options given below:

(1) A-II, B-III, C-I, D-IV
(2) A-II, B-III, C-IV, D-I
(3) A-IV, B-III, C-II, D-I
(4) A-III, B-II, C-IV, D-I

118. Given below are two statements:

Statement I: Mineralocorticoid deficiency greatly decreases renal tubular sodium reabsorption and consequently allows sodium ions, chloride ions, and water to be lost into urine in great quantity.

Statement II: Cortisol suppresses the immune system, especially the T-lymphocyte production to avoid graft rejection.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I and Statement II both are correct.
(2) Statement I and Statement II both are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

119. Using a prism, T.W. Engelmann split light into its spectral components and then illuminated a:

(1) mint plant.
(2) green alga.
(3) purple bacterium.
(4) green sulphur bacterium.

120. Under what condition is a fruit classified as a true fruit?

(1) If ovary of the flower develops into fruit.
(2) When calyx of the flower develops into fruit.
(3) When thalamus of the flower develops into fruit.
(4) If all floral whorls of the flower develop into fruit.

121. Choose the **incorrect** statement w.r.t effect of temperature on spermatogenesis in human males.

(1) The metabolic processes and enzymes involved in meiosis and spermatogenesis are more efficient at lower temperature i.e. approximately 34°C to 35°C.
(2) Testes are located in scrotal sacs to maintain the temperature of seminiferous tubules below the internal temperature of the body.
(3) The scrotum acts as a heating device for the testes, without which spermatogenesis might be inefficient.
(4) On cold days, scrotal reflexes cause the musculature of the scrotum to contract, pulling the testes close to the body to maintain appropriate temperature.

122. Who observed that within a region, species richness increased with increasing explored area but only upto a limit?

(1) Paul Ehrlich
(2) David Tilman
(3) Alexander von Humboldt
(4) Edward Wilson

123. Given below are two statements:

Statement-I: The strand of DNA which does not code for anything is referred to as coding strand.

Statement-II: The promoter and terminator flank the structural gene in a transcription unit.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

124. Nicotinamide adenine dinucleotide (NAD) contains the vitamin niacin. Here

(1) Niacin acts as prosthetic group
(2) NAD acts as coenzyme
(3) NADP is tightly bound to the apoenzyme
(4) Niacin is apoenzyme

125. _____, component of innate immunity are given to cancer patients to help in destroying the tumor. Select the **correct** option to fill in the blank.

(1) Insulin-like growth factors
(2) α-interferons
(3) α-lactalbumin
(4) α-antitrypsin

[Contd...

126. How many of the following are **correct** statements w.r.t. human excretory system?

A. All the structures associated with it are paired except urinary bladder.
B. Kidneys are bean-shaped structures situated between last thoracic and third lumbar vertebra.
C. Kidneys are situated close to ventral inner wall of the abdominal cavity
D. Projections of renal capsule towards hilum are called renal calyces.
E. Renal medulla is divided into few medullary pyramids projecting into calyces.

Choose the **correct** option.

(1) Five (2) Three
(3) Four (4) Two

127. Select the **incorrectly** matched pair from the following.

(1) Zeatin - Isolated from corn kernels and coconut milk
(2) Abscisic acid - Isolated from ripened oranges
(3) Gibberellins - Isolated from the fungus *Gibberella fujikuroi*
(4) Auxin - Isolated from human urine

128. Read the processes given below and choose the **correct** option where Coenzyme A is required as a substrate.

A. Glycolysis
B. Lactic acid fermentation
C. Krebs' cycle
D. Oxidative decarboxylation of pyruvate

Choose the *most appropriate* answer from the options given below:

(1) C and D only (2) B only
(3) C only (4) A and C only

129. If there is only one PMC in a microsporangium, then the total number of male gametes produced by it is:

(1) four only.
(2) two only.
(3) eight.
(4) six only.

130. All of the following processes are involved in the secondary treatment of sewage **except:**

(1) aerobic digestion of organic matter.
(2) BOD reduction and biogas formation.
(3) sequential filtration and sedimentation.
(4) anaerobic digestion of bacteria and fungi.

131. Consider the following statements.

A. Plants, bacteria, fungi and animals whose genes have been altered by manipulation, are called GMOs.
B. Genetic modification has been used to create tailor-made plants to supply alternative resources to industries
C. *Bacillus thuringiensis* form protein crystals throughout their life, which contain a toxic insecticidal protein.
D. Bt toxin protein exists as an inactive protoxin but once ingested by an insect, it is converted into its active form due to acidic pH of the insect gut.

Which of the above statements are **not** true?

(1) A and B (2) B and C
(3) C and D (4) A and D

132. The sac-like structure which receives blood from ventricle of frog is known as

(1) Sinus venosus
(2) Vena cava
(3) Conus arteriosus
(4) Portal vein

133. A laboratory technician places a drop of blood from a volunteer onto three separate petri dishes labelled as P, Q and R.

Each dish contains anti-serum as follows:

- Petri dish P : Anti-A serum
- Petri dish Q : Anti-B serum
- Petri dish R : Anti-Rh serum

After mixing, the following observations are recorded:

A. Strong clumping appears only in dishes P and R.
B. No clumping in dish Q.

The **correct** blood group of the volunteer is

(1) A^{-ve}
(2) A^{+ve}
(3) AB^{+ve}
(4) O^{+ve}

134. The rate of biomass production is expressed in terms of:

(1) kcal m.
(2) kcal m^{-2} yr.
(3) g m^{-2}.
(4) g m^{-2} yr^{-1}.

135. In angiosperms, the ovules:

(1) contains microsporangium.
(2) may be either one or many in each ovary.
(3) lack nucellus in seeds of beet.
(4) have an integument at micropylar end.

[Contd...

136. The restriction endonuclease enzymes belong to the class of enzymes and act on the bond that joins the sugar residues of adjacent nucleotides within the backbone of double-stranded DNA.
Select the **correct** option to fill the respective blanks.
(1) Ligases; Hydrogen
(2) Nucleases; Phosphodiester
(3) Oxidoreductases; Hydrogen
(4) Isomerase; Phosphodiester

137. Actinomorphic symmetry is present in the flowers of all **except:**
(1) chilli.
(2) mustard.
(3) pea.
(4) *Datura*.

138. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Cyclosporin A	(I)	*Clostridium butylicum*
(B)	Statin	(II)	*Streptococcus*
(C)	Streptokinase	(III)	*Monascus purpureus*
(D)	Butyric acid	(IV)	*Trichoderma polysporum*

Choose the **correct** answer from the options given below:
(1) A-III, B-II, C-I, D-IV
(2) A-IV, B-III, C-II, D-I
(3) A-II, B-III, C-IV, D-I
(4) A-II, B-I, C-III, D-IV

139. Match the **List-I** with **List-II**.

List-I		List-II	
(A)	Francois Jacob and Jacque Monod	(I)	DNA fingerprinting
(B)	Alec Jeffreys	(II)	*lac* operon
(C)	Frederick Griffith	(III)	Proposed double helical structure for DNA
(D)	James Watson and Francis Crick	(IV)	Transforming principle

Choose the **correct** answer from the options given below:
(1) A-II, B-IV, C-I, D-III
(2) A-III, B-IV, C-II, D-I
(3) A-II, B-I, C-IV, D-III
(4) A-I, B-III, C-IV, D-II

140. Transgenic animals are not produced for which of the following purposes?
(1) To study how genes are regulated and how they affect the normal functions of body and its development.
(2) To study the diseases.
(3) To test vaccine safety and chemical safety.
(4) To obtain harmful products.

141. Match List-I with List-II.

List-I		List-II	
(A)	Pachytene	(I)	Terminalisation of chiasmata
(B)	Diakinesis	(II)	Crossing over and recombination
(C)	Leptotene	(III)	Chromosome synapsis
(D)	Zygotene	(IV)	Gradual visibility of chromosomes under a light microscope

Choose the **correct** answer from the options given below:
(1) A-I, B-II, C-III, D-IV
(2) A-I, B-III, C-II, D-IV
(3) A-II, B-I, C-IV, D-III
(4) A-IV, B-I, C-II, D-III

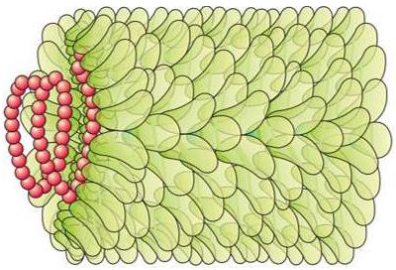
142. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Cerebrum	(I)	Regulation of respiration and cardiovascular reflexes
(B)	Cerebellum	(II)	Maintenance of posture and balance
(C)	Medulla Oblongata	(III)	Centre for intelligence, memory and voluntary actions
(D)	Hypothalamus	(IV)	Regulation of body temperature and hunger

Choose the **correct** answer from the options given below:
(1) A-III, B-II, C-I, D-IV
(2) A-II, B-IV, C-IV, D-I
(3) A-III, B-I, C-II, D-IV
(4) A-IV, B-II, C-I, D-III

[Contd...

143. The genetic material found in the following organism is:



(1) DNA only.
(2) RNA only.
(3) protein only.
(4) DNA, RNA and protein.

144. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Herbivores and plants	(I)	Commensalism
(B)	Mycorrhiza	(II)	Mutualism
(C)	Flamingoes and fishes	(III)	Predation
(D)	Orchid and mango tree	(IV)	Competition

Choose the **correct** answer from the options given below:
(1) A-III, B-II, C-IV, D-I
(2) A-II, B-I, C-III, D-IV
(3) A-I, B-III, C-IV, D-II
(4) A-IV, B-II, C-I, D-III

145. The human liver fluke, a trematode parasite depends on two intermediate hosts ___**A**___ and ____**B**___ to complete its cycle. Identify **A** and **B** respectively.
(1) A - human, B - snail
(2) A - fish, B - human
(3) A - snail, B - fish
(4) A - snail, B - mosquito

146. The **correct** restriction site where *Eco*RI acts is represented by

(1) 3′ C T T A A G 5′
5′ G A A T T C 3′

(2) 5′ C T T A A G 3′
3′ G A A T T C 5′

(3) 5′ G A A T T C 5′
3′ C T T A A G 3′

(4) 5′ G A A T T C 3′
3′ C T T A A G 5′

147. Consider the following statements regarding cancer:
A. Cancer is characterized by uncontrolled proliferation of cells.
B. Benign tumors always invade surrounding tissues and metastasize.
C. Carcinogens include certain chemicals, radiations, and oncogenic viruses.
D. Normal cells can transform into cancerous cells due to genetic mutations.
E. Metastasis refers to the spread of cancer cells from the primary site to distant organs.
How many of the above statements are correct?
(1) 2 (2) 3
(3) 4 (4) 5

148. Select the **incorrect** match w.r.t a sarcomere.
(1) 'H' zone — Central part of thick filament, not overlapped by thin filaments
(2) 'A' band — Possesses both actin and myosin proteins
(3) 'I' band — Contains the protein which is made of two 'F' actins helically wound to each other
(4) 'Z' line — Inelastic fibre in the centre of each 'I' band

149. A typical bacterial cell **lacks** which of the following?
A. Cell wall B. Cilia
C. Flagella D. Circular DNA
E. 70S ribosomes
Choose the *most appropriate* answer from the options given below:
(1) A, B and E only
(2) A and C only
(3) B only
(4) B and D only

150. Match the level of coelom with the Phylum:

List-I		List-II	
(A)	Acoelomate	(I)	Aschelminthes
(B)	Pseudocoelomate	(II)	Platyhelminthes
(C)	Coelomate	(III)	Annelida
(D)	Cellular aggregates	(IV)	Porifera

Choose the **correct** answer from the options given below:
(1) A-III, B-I, C-II, D-IV
(2) A-I, B-II, C-III, D-IV
(3) A-II, B-I, C-III, D-IV
(4) A-IV, B-III, C-II, D-I

151. Which of the given algae have stored food that is very similar to amylopectin and glycogen in structure?
(1) *Chara*
(2) *Volvox*
(3) *Fucus*
(4) *Porphyra*

[Contd...

152. Select the **correct** match w.r.t. location of different endocrine glands in humans.
(1) Pineal gland - Ventral side of forebrain
(2) Thymus gland - Dorsal side of aorta
(3) Adrenal gland - Lateral surface of kidney
(4) Pituitary gland - Inside bony cavity called sella tursica of sphenoid bone

153. Select the **correct** statement/s w.r.t. primary productivity.
A. It depends on the plant species inhabiting a particular area.
B. It depends on photosynthetic capacity of plants.
C. It is same in all types of ecosystems.
D. For a natural ecosystem it depends upon availability of nutrients in soil.
Choose the *most appropriate* answer from the options given below:
(1) A and B only
(2) A, B and D only
(3) B, C and D only
(4) C and D only

154. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Golgi apparatus	(I)	Ribosomal RNA
(B)	Nucleolus	(II)	Form glycoproteins and glycolipids
(C)	Cilia	(III)	Helps in osmoregulation and excretion
(D)	Contractile vacuole	(IV)	Microtubules

Choose the **correct** answer from the options given below.
(1) A-I, B-II, C-III, D-IV
(2) A-IV, B-III, C-II, D-I
(3) A-II, B-I, C-IV, D-III
(4) A-III, B-IV, C-I, D-II

155. Identify the **correct** statements regarding binomial nomenclature.
(1) Name of the author appears before the specific epithet.
(2) Both genus and specific epithet are printed in italics.
(3) The specific epithet starts with a capital letter.
(4) The first word represents the specific epithet and second component denotes the genus.

156. A chain of 10 glucose molecules will have how many peptide bonds?
(1) Nine (2) Zero
(3) Eight (4) Ten

157. Which of the following is a non-hormonal contraceptive?
(1) LNG-20
(2) Multiload-375
(3) Implants
(4) Oral pills

158. Given below are two statements:
Statement I: Respiratory pigment, haemoglobin, in humans contains 4 different subunits.
Statement II: Tertiary structure which gives us a 3-dimensional view of a protein is absolutely necessary for the many biological activities of proteins.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I and Statement II both are correct.
(2) Statement I and Statement II both are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

159. The digestive system in the members of the phylum Platyhelminthes is incomplete because their digestive system has
(1) No opening for ingestion and egestion
(2) Well developed muscular pharynx
(3) Separate openings for mouth and anus
(4) Only a single opening to the outside that serves as both mouth and anus.

160. Match **List-I** with **List-II**.

List-I (Endocrine Gland / Hormone)		List-II (Function)	
(A)	Pituitary gland	(I)	Regulates basal metabolic rate
(B)	Thyroxine	(II)	Releases adrenaline during emergency
(C)	Adrenal medulla	(III)	Regulates blood glucose level
(D)	Pancreas	(IV)	Master gland controlling other endocrine glands

Choose the **correct** answer from the options given below:
(1) A-IV, B-I, C-II, D-III
(2) A-I, B-IV, C-II, D-III
(3) A-IV, B-II, C-I, D-III
(4) A-III, B-I, C-IV, D-II

161. According to the Rivet Popper Hypothesis, which of the following is **not** comparable?
(1) Safety of flight - Stability of ecosystem
(2) Rivets - Species
(3) Airplane - Ecosystem
(4) Popping of rivets - Origin of species

162. Choose the parts of renal tubules which participate in maintaining ionic balance as well as pH of the body fluids by secreting H^+ and ammonia in filtrate.
(1) PCT and descending limb of loop of Henle
(2) PCT and thin ascending limb of loop of Henle
(3) DCT and collecting duct
(4) PCT and DCT

163. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion (A): Sterilization is the permanent prevention of gamete and hormone release.
Reason (R): Sterilization neither affect sexual arousal nor menstrual cycle of an individual.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is NOT the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

164. In the respiratory pathway, glycerol would enter after being converted to:
(1) 3-phosphoglyceraldehyde (PGAL).
(2) 1,3-bisphosphoglycerate (BPGA).
(3) phosphoenolpyruvate.
(4) acetyl CoA.

165. In the process of translation, the order and sequence of amino acids are defined by the:
(1) sequence of bases in the mRNA.
(2) sequence of bases in the rRNA.
(3) sequence of bases in the tRNA.
(4) sequence of bases in the UTRs.

166. Based on observations made during a sea voyage, who among the following concluded that existing living forms share similarities to varying degrees not only among themselves but also with life forms that existed millions of years ago?
(1) Lamarck (2) Charles Darwin
(3) Von Baer (4) Ernst Heckel

167. Read the following statements w.r.t. the human respiratory system.
Statement I: Each bronchi undergoes repeated divisions to form the secondary and tertiary bronchi and bronchioles.
Statement II: Only trachea and the primary bronchi are supported by cartilaginous rings.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I and Statement II both are correct.
(2) Statement I and Statement II both are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

168. In a monocot seed, the plumule and radicle are enclosed in sheaths which are called ___**A**___ and ___**B**___ respectively. Identify **A** and **B**.
(1) A-coleoptile, B-coleorhiza
(2) A-coleorhiza, B-coleoptile
(3) A-aleurone layer, B-scutellum
(4) A-scutellum, B-aleurone layer

169. Select the **incorrect** match among the following w.r.t. body symmetry.
(1) Adult *Echinus* - Radial
(2) *Laccifer* - Radial
(3) *Saccoglossus* - Bilateral
(4) *Octopus* - Bilateral

170. For performing an experiment, a scientist wanted to isolate the genetic material of the organism 'A' that causes Typhoid and the organism 'B' that causes ringworms.
He began the process of isolation in his lab by adding enzymes in a test tube.
From the options given below, Choose the **correct** sets of enzymes that should be added in the respective tubes.
[Note : Tube I → Contains organism 'B'
Tube II → Contains organism 'A']
(1) Tube I → Ribonuclease + DNase + Protease
Tube II → Lysozyme + RNase + Protease
(2) Tube I → Protease + Lysozyme + Chitinase
Tube II → Ribonuclease + Protease + Lysozyme
(3) Tube I → Lysozyme + Protease + Ribonuclease
Tube II → Protease + Chitinase + Ribonuclease
(4) Tube I → Chitinase + Protease + Ribonuclease
Tube II → Lysozyme + Protease + Ribonuclease

171. Read the following functions of the plant growth regulators (PGRs).
I. Bolting in a rosette plant
II. Promote seed dormancy
III. Flowering in mango tree
Choose the correct PGR used for **I, II and III**.
(1) I - Gibberellins, II - Abscisic acid and III - Cytokinin
(2) I - Auxin, II - Cytokinin and III - Ethylene
(3) I - Auxin, II - Cytokinin and III - Gibberellins
(4) I - Gibberellins, II - Abscisic acid and III - Ethylene

172. Given below are two statements:
Statement I: The pleural fluid present in pleural cavity reduces friction between pleural membranes.
Statement II: The concave, medial surface of each lung contains a region through which bronchi, pulmonary blood vessels enter.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I and Statement II both are correct.
(2) Statement I and Statement II both are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

173. Telophase II in meiosis ends with the formation of:
(1) tetrad of haploid cells.
(2) tetrad of diploid cells.
(3) dyad of diploid cells.
(4) dyad of haploid cells.

174. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: When alien species are introduced unintentionally some of them cause extinction of indigenous species.
Reason R: Introduction of *Clarias gariepinus* in Indian rivers has led to a decline in native Indian plants.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

175. Identify the **correctly** matched pair in C_4 plants from the following.
(1) Bundle sheath cells-RuBisCO present
(2) Mesophyll cells-PEPcase absent
(3) Mesophyll cells- RuBisCO present
(4) Bundle sheath cells-PEPcase present

176. The major pigments found in brown algae are:
(1) r-phycoerythrin.
(2) chlorophyll *a* and *b*.
(3) chlorophyll *a*, *c* and fucoxanthin.
(4) chlorophyll *b*, *d* and carotenoids.

177. Which of the following pairs is **incorrectly** matched?
(1) *Entamoeba* – Protozoa
(2) *Nostoc* – Cyanobacteria
(3) *Colletotrichum* – Chrysophytes
(4) *Gonyaulax* – Dinoflagellates

178. Read the following statements and select how many of them are **incorrect**?
A. Conjoint type of vascular bundles is seen in leaves.
B. Trichomes are found in the stem and help in preventing water loss due to transpiration.
C. Epidermis is usually multilayered in dicot roots.
D. The cells of epidermis in roots bear a number of unicellular hairs.
Choose the *most appropriate* answer from the options given below:
(1) One only
(2) Two only
(3) Three only
(4) Four only

179. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The specific epithet of the lion is *leo*.
Reason R: The lion belongs to family Canidae.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

180. Which of the following is an example of evolution due to anthropogenic action but is not an example of natural selection?
(1) Herbicide resistant weeds
(2) Pesticide resistant pests
(3) New breeds of dogs produced by intensive breeding programmes
(4) Antibiotic resistant microbes

■■■

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